# Run report: ctrlH-q3-run-20261024T2151Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part HYPER under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations ($F_p$ multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

## env.txt

```
run: ctrlH-q3 20261024T2151Z
git HEAD: 0fb870d0ee76ba6a1b059e9e34c869518fb412b0
tool: examples/genus4_n4_census.rs sha256 f4c7f7355c838b6f112e09cb412fd3353d087309962c64cd39ad49d79f4c9dad
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 a95b1e35748cefe23a044d8cfd9e059b723cf71fdbc4af56db71257da42c8b33
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/ctrlH-q3-run-20261024T2151Z HYPER 3 exh 1
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 0
```

## tool_log.txt

```
HYPER q=3 mode=exhaustive forms_examined=177147
squarefree_degree10_forms=104976
res_shaped=1100
odd_t_hits=0
t_hist={-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
seconds=0.1
```

## summary.txt

```
HYPER q=3 mode=exhaustive forms_examined=177147
squarefree_degree10_forms=104976
res_shaped=1100
odd_t_hits=0
t_hist={-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
seconds=0.1
```

## hyper_t_hist.csv (first 12 of 12 rows)

| t | count |
|---|---|
| −18 | 18 |
| −10 | 96 |
| −8 | 144 |
| −6 | 96 |
| −2 | 192 |
| 0 | 204 |
| 2 | 180 |
| 4 | 48 |
| 8 | 48 |
| 10 | 48 |
| 16 | 24 |
| 18 | 2 |

**Graph**

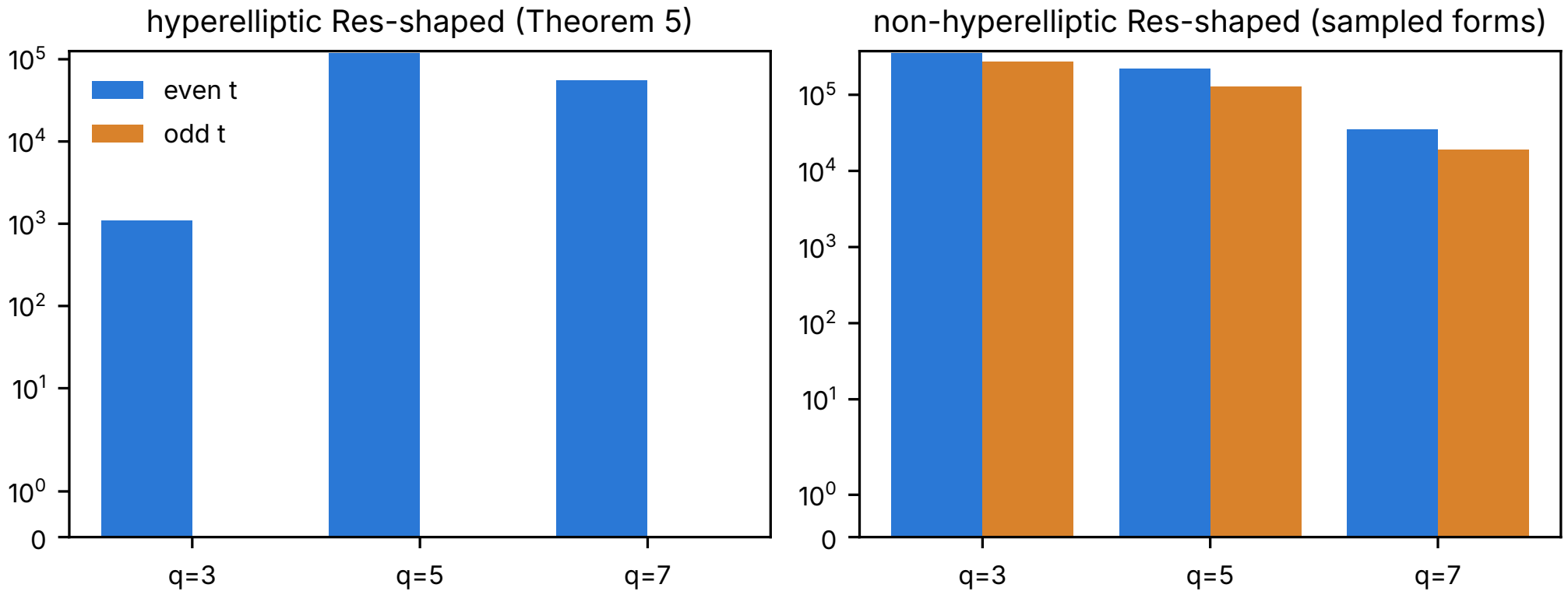


Figure 1: Parity of Res-shaped traces: hyperelliptic (Theorem 5, no odd t) and non-hyperelliptic.